

## Problem

You are given an array of N integers representing the weights of people.

Your task is to divide the people into two groups such that:

The two groups have the same number of people, or their sizes differ by at most one. The absolute difference between the total weights of the two groups is minimized.

Print the minimum possible difference between the total weights of the two groups.

## Submissions

## Input Format

The first line contains an integer N, the number of people. The second line contains N space-separated integers representing their weights.

## Constraints

 $2 \leq N \leq 30$   $1 \leq arr[i] \leq 1000$ 

## Output Format

Print a single integer representing the minimum possible difference between the total weights of the two groups.

## Leaderboard

## Discussions

## Sample Input 0

4  
3 4 5 6

## Sample Output 0

0

## Sample Input 1

5  
1 2 3 4 5

## Sample Output 1

1

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Language Java 15



```
1  import java.util.*;
2
3  public class Main {
4
5      static int n;
6      static int[] arr;
7      static int total;
8      static int minDiff = Integer.MAX_VALUE;
9
10     static void solve(int index, int count, int sum) {
11
12         if (index == n) {
13             int group2 = total - sum;
14
15             if (Math.abs(count - (n - count)) < 2) {
16                 int diff = Math.abs(sum - group2);
17                 minDiff = Math.min(minDiff, diff);
18             }
19         }
20         return;
21     }
22
23     // Include current person
24     solve(index + 1, count + 1, sum + arr[index]);
25 }
```

Line: 47 Col: 2



Upload Code as File

Run Code

Submit Code



Test against custom input

## Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0



Test case 1

Compiler Message

Success